

17.2 Multi-period Forecasts

The discussion of forecasting so far has focused on making forecasts one period in advance. Often, however, forecasters are called upon to make forecasts further into the future. This section describes two methods for making multi-period forecasts, which are also called multi-step forecasts. The first method is to construct iterated forecasts, in which a one-period ahead model is iterated forward one period at a time in a way that is made precise in this section. The second method is to make direct forecasts by using a regression in which the dependent variable is the multi-period variable that one wants to forecast. For reasons discussed at the end of this section, in most applications the iterated method is recommended over the direct method.

Iterated Multi-period Forecasts

The essential idea of an iterated forecast is that a forecasting model is used to make a forecast one period ahead, for period $T + 1$, using data through period T . The model then is used to make a forecast for date $T + 2$, given the data through date T , where the forecasted value for date $T + 1$ is treated as data for the purpose of making the forecast for period $T + 2$. Thus the one-period ahead forecast (which is also referred to as a one-step ahead forecast) is used as an intermediate step to make the two-period ahead forecast. This process repeats, or iterates, until the forecast is made for the desired forecast horizon h .

The iterated AR forecast method: AR(1). An iterated AR(1) forecast uses an AR(1) for the one-period ahead model. For example, consider the first-order autoregression for $GDPGR$ [Equation (15.9)]:

$$\widehat{GDPGR}_t = 1.95 + 0.34 GDPGR_{t-1}. \quad (17.7)$$

(0.32) (0.07)

The first step in computing the two-quarter ahead forecast of $GDPGR_{2018:Q1}$ based on Equation (17.7) and using data through 2017:Q3 is to compute the one-quarter ahead forecast of $GDPGR_{2017:Q4}$ based on data through 2017:Q3: $\widehat{GDPGR}_{2017:Q4|2017:Q3} = 1.95 + 0.34 GDPGR_{2017:Q3} = 1.95 + 0.34 \times 3.11 = 3.0$. The second step is to substitute this forecast into Equation (17.7), so that $\widehat{GDPGR}_{2018:Q1|2017:Q3} = 1.95 + 0.34 \widehat{GDPGR}_{2017:Q4|2017:Q3} = 1.95 + 0.34 \times 3.0 = 3.0$. Thus, based on information through the third quarter of 2017, this forecast states that the growth rate of GDP will be 3.0% in the first quarter of 2018.

The iterated AR forecast method: AR(p). The iterated AR(1) strategy is extended to an AR(p) by replacing Y_{T+1} with its forecast, $\hat{Y}_{T+1|T}$, and then treating that forecast as data for the AR(p) forecast of Y_{T+2} . For example, consider the iterated two-period

ahead forecast of the growth rate of GDP based on the AR(2) model from Section 15.3 [Equation (15.11)]:

$$\widehat{GDPGR}_t = 1.60 + 0.28 \widehat{GDPGR}_{t-1} + 0.18 \widehat{GDPGR}_{t-2}. \quad (17.8)$$

(0.37) (0.08) (0.08)

The forecast of $GDPGR_{2017:Q4}$ based on data through 2017:Q3 using this AR(2), computed in Section 15.3, is $\widehat{GDPGR}_{2017:Q4|2017:Q3} = 3.0$. Thus the two-quarter ahead iterated forecast based on the AR(2) is $\widehat{GDPGR}_{2018:Q1|2017:Q3} = 1.60 + 0.28 \widehat{GDPGR}_{2017:Q4|2017:Q3} + 0.18 \widehat{GDPGR}_{2017:Q3} = 1.60 + 0.28 \times 3.0 + 0.18 \times 3.1 = 3.0$. According to this iterated AR(2) forecast, based on data through the third quarter of 2017, the growth rate of GDP is predicted to be 3.0% in the first quarter of 2018.

Iterated multivariate forecasts using an iterated VAR. Iterated multivariate forecasts can be computed using a VAR in much the same way as iterated univariate forecasts are computed using an autoregression. The main new feature of an iterated multivariate forecast is that the two-step ahead (period $T + 2$) forecast of one variable depends on the forecasts of all variables in the VAR in period $T + 1$. For example, to compute the forecast of the growth rate of GDP in period $T + 2$ using a VAR with the variables $GDPGR_t$ and $TSpread_t$, one must forecast both $GDPGR_{T+1}$ and $TSpread_{T+1}$, using data through period T as an intermediate step in forecasting $GDPGR_{T+2}$. More generally, to compute multi-period iterated VAR forecasts h periods ahead, it is necessary to compute forecasts of all variables for all intervening periods between T and $T + h$.

As an example, we will compute the iterated VAR forecast of $GDPGR_{2018:Q1}$ based on data through 2017:Q3, using the VAR(2) for $GDPGR_t$ and $TSpread_t$ in Section 17.1 [Equations (17.5) and (17.6)]. The first step is to compute the one-quarter ahead forecasts $\widehat{GDPGR}_{2017:Q4|2017:Q3}$ and $\widehat{TSpread}_{2017:Q4|2017:Q3}$ from that VAR. These one-period ahead forecasts were computed in Section 17.1 based on Equations (17.5) and (17.6). The forecasts were $\widehat{GDPGR}_{2017:Q4|2017:Q3} = 2.8$ and $\widehat{TSpread}_{2017:Q4|2017:Q3} = 1.3$. In the second step, these forecasts are substituted into Equations (17.5) and (17.6) to produce the two-quarter ahead forecast:

$$\begin{aligned} \widehat{GDPGR}_{2018:Q1|2017:Q3} &= 0.54 + 0.29 \widehat{GDPGR}_{2017:Q4|2017:Q3} + 0.20 \widehat{GDPGR}_{2017:Q3} \\ &\quad - 0.86 \widehat{TSpread}_{2017:Q4|2017:Q3} + 1.28 \widehat{TSpread}_{2017:Q3} \\ &= 0.54 + 0.29 \times 2.8 + 0.20 \times 3.1 \\ &\quad - 0.86 \times 1.3 + 1.28 \times 1.2 = 2.4. \end{aligned} \quad (17.9)$$

Thus the iterated VAR(2) forecast, based on data through the third quarter of 2017, is that the growth rate of GDP will be 2.4% in the first quarter of 2018.

Iterated multi-period forecasts are summarized in Key Concept 17.2.

KEY CONCEPT

Iterated Multi-period Forecasts

17.2

The **iterated multi-period AR forecast** is computed in steps: First compute the one-period ahead forecast, and then use that to compute the two-period ahead forecast, and so forth. The two- and three-period ahead iterated forecasts based on an AR(p) are

$$\hat{Y}_{T+2|T} = \hat{\beta}_0 + \hat{\beta}_1 \hat{Y}_{T+1|T} + \hat{\beta}_2 Y_T + \hat{\beta}_3 Y_{T+1} + \cdots + \hat{\beta}_p Y_{T-p+2} \quad (17.10)$$

$$\hat{Y}_{T+3|T} = \hat{\beta}_0 + \hat{\beta}_1 \hat{Y}_{T+2|T} + \hat{\beta}_2 \hat{Y}_{T+1|T} + \hat{\beta}_3 Y_T + \cdots + \hat{\beta}_p Y_{T-p+3}, \quad (17.11)$$

where the $\hat{\beta}$'s are the OLS estimates of the AR(p) coefficients. Continuing this process (iterating) produces forecasts further into the future.

The **iterated multi-period VAR forecast** is also computed in steps: First compute the one-period ahead forecast of all the variables in the VAR, then use those forecasts to compute the two-period ahead forecasts, and continue this process iteratively to the desired forecast horizon. The two-period ahead iterated forecast of Y_{T+2} , based on the two-variable VAR(p) in Key Concept 17.1, is

$$\begin{aligned} \hat{Y}_{T+2|T} = & \hat{\beta}_{10} + \hat{\beta}_{11} \hat{Y}_{T+1|T} + \hat{\beta}_{12} Y_T + \hat{\beta}_{13} Y_{T-1} + \cdots + \hat{\beta}_{1p} Y_{T-p+2} \\ & + \hat{\gamma}_{11} \hat{X}_{T+1|T} + \hat{\gamma}_{12} X_T + \hat{\gamma}_{13} X_{T-1} + \cdots + \hat{\gamma}_{1p} X_{T-p+2}, \end{aligned} \quad (17.12)$$

where the coefficients in Equation (17.12) are the OLS estimates of the VAR coefficients. Iterating produces forecasts further into the future.

Direct Multi-period Forecasts

Direct multi-period forecasts are computed without iterating by using a single regression, in which the dependent variable is the multi-period ahead variable to be forecasted and the regressors are the predictor variables. Forecasts computed this way are called direct forecasts because the regression coefficients can be used directly to make the multi-period forecast.

The direct multi-period forecasting method. Suppose that you want to make a forecast of Y_{T+2} using data through time T . The direct multivariate method takes the ADL model as its starting point but lags the predictor variables by an additional time period. For example, if two lags of the predictors are used, then the dependent variable is Y_t and the regressors are Y_{t-2} , Y_{t-3} , X_{t-2} , and X_{t-3} . The coefficients from this regression can be used directly to compute the forecast of Y_{T+2} using data on Y_T , Y_{T-1} , X_T , and X_{T-1} , without the need for any iteration. More generally, in a direct h -period ahead forecasting regression, all predictors are lagged h periods to produce the h -period ahead forecast.

For example, the forecast of $GDPGR_t$ two quarters ahead using two lags each of $GDPGR_{t-2}$ and $TSpread_{t-2}$ is computed by first estimating the regression:

$$\begin{aligned} \widehat{GDPGR}_{t|t-2} = & 0.56 + 0.31GDPGR_{t-2} + 0.04GDPGR_{t-3} \\ & (0.63) \quad (0.07) \quad (0.09) \\ & + 0.56TSpread_{t-2} + 0.04TSpread_{t-3}. \\ & (0.46) \quad (0.45) \end{aligned} \quad (17.13)$$

The two-quarter ahead forecast of the growth rate of GDP in 2018:Q1 based on data through 2017:Q3 is computed by substituting the values of $GDPGR_{2017:Q3}$, $GDPGR_{2017:Q2}$, $TSpread_{2017:Q3}$, and $TSpread_{2017:Q2}$ into Equation (17.13); this yields

$$\begin{aligned} \widehat{GDPGR}_{2018:Q1|2017:Q3} = & 0.56 + 0.31GDPGR_{2017:Q3} + 0.04GDPGR_{2017:Q2} \\ & + 0.56TSpread_{2017:Q3} + 0.04TSpread_{2017:Q2} = 2.4. \end{aligned} \quad (17.14)$$

The three-quarter ahead direct forecast of $GDPGR_{T+3}$ is computed by lagging all the regressors in Equation (17.13) by one additional quarter, estimating that regression, and then computing the forecast. The h -quarter ahead direct forecast of $GDPGR_{T+h}$ is computed by using $GDPGR_t$ as the dependent variable and the regressors $GDPGR_{t-h}$ and $TSpread_{t-h}$ plus additional lags of $GDPGR_{t-h}$ and $TSpread_{t-h}$, as desired.

Standard errors in direct multi-period regressions. Because the dependent variable in a multi-period regression occurs two or more periods into the future, the error term in a multi-period regression is serially correlated. To see this, consider the two-period ahead forecast of the GDP growth rate, and suppose that a surprise jump in oil prices occurs in the next quarter. Today's two-period ahead forecast of the growth rate of GDP will be too high because it does not incorporate this unexpected negative event. Because the oil price rise was also unknown in the previous quarter, the two-period ahead forecast made last quarter will also be too high. Thus the surprise oil price jump next quarter means that *both* last quarter's and this quarter's two-period ahead forecasts are too high. Because of such intervening events, the error term in a multi-period regression is serially correlated.

As discussed in Section 16.4, if the error term is serially correlated, the usual OLS standard errors are incorrect, or, more precisely, they are not a reliable basis for inference. Therefore, heteroskedasticity- and autocorrelation-consistent (HAC) standard errors must be used with direct multi-period regressions. The standard errors reported in Equation (17.13) for direct multi-period regressions therefore are Newey–West HAC standard errors, where the truncation parameter m is set according to Equation (16.17); for these data (for which $T = 147$), Equation (16.17) yields $m = 4$. For longer forecast horizons, the amount of overlap—and thus the degree of serial correlation in the error—increases: In general, the first $h - 1$ autocorrelation coefficients of the errors in an h -period ahead regression are nonzero. Thus larger values of m than indicated by Equation (16.17) are appropriate for multi-period regressions with long forecast horizons.

Direct multi-period forecasts are summarized in Key Concept 17.3.

KEY CONCEPT

Direct Multi-period Forecasts

17.3

The **direct multi-period forecast** h periods into the future based on p lags each of Y_t and an additional predictor X_t is computed by first estimating the regression

$$Y_t = \delta_0 + \delta_1 Y_{t-h} + \cdots + \delta_p Y_{t-p-h+1} + \delta_{p+1} X_{t-h} + \cdots + \delta_{2p} X_{t-p-h+1} + u_t \quad (17.15)$$

and then using the estimated coefficients directly to make the forecast of Y_{T+h} using data through period T .

Which Method Should You Use?

In most applications, the iterated method is the recommended procedure for multi-period forecasting for two reasons. First, from a theoretical perspective, if the underlying one-period ahead model (the AR or VAR that is used to compute the iterated forecast) is specified correctly, then the coefficients are estimated more efficiently if they are estimated by a one-period ahead regression (and then iterated) than by a multi-period ahead regression. Second, from a practical perspective, forecasters are usually interested in forecasts not just at a single horizon but at multiple horizons. Because they are produced using the same model, iterated forecasts tend to have time paths that are less erratic across horizons than do direct forecasts. Because a different model is used at every horizon for direct forecasts, sampling error in the estimated coefficients can add random fluctuations to the time paths of a sequence of direct multi-period forecasts.

Under some circumstances, however, direct forecasts are preferable to iterated forecasts. One such circumstance is when you have reason to believe that the one-period ahead model (the AR or VAR) is not specified correctly. For example, you might believe that the equation for the variable you are trying to forecast in a VAR is specified correctly but that one or more of the other equations in the VAR are specified incorrectly, perhaps because of neglected nonlinear terms. If the one-step ahead model is specified incorrectly, then, in general, the iterated multi-period forecast will be biased, and the MSFE of the iterated forecast can exceed the MSFE of the direct forecast, even though the direct forecast has a larger variance.

17.3 Orders of Integration and the Nonnormality of Unit Root Test Statistics

This section extends the treatment of stochastic trends in Section 15.6 by addressing two further topics. First, the trends of some time series are not well described by the random walk model, so we introduce an extension of that model and discuss its